Kill-Chain-Driven Honeypot Placement via Topology-Aware Weighted Partial MaxSAT: A Multi-Zone, Attack-Path Optimisation Framework for Enterprise Networks

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***Abstract*—Deciding where to place honeypots in a large enterprise network is NP-hard. The network is split into security zones, contains thousands of server types, and faces attackers following structured kill-chain patterns. Budget constraints and pairwise honeypot conflicts further restrict feasible deployments.**

We model honeypot placement as a Weighted Partial Maximum Satisfiability (MaxSAT) problem with six simultaneous objectives: detection efficiency, MITRE ATT&CK technique coverage, tactic-family breadth, early attacker interception, forensic backward coverage, and multi-path coverage. All hard rules (budget, zone isolation, conflict constraints) are encoded as inviolable clauses. We prove NP-hardness via polynomial-time reduction from Weighted Maximum Coverage and show that a cross-path force-multiplier—where one honeypot satisfies clauses for multiple attack paths simultaneously—gives our solver $2.1\times$ more weight per decision than any greedy method.

We evaluate on six network scales (100 to 5.5 million nodes) and validate on a Mininet/OpenFlow testbed with real Cowrie SSH and Conpot ICS honeypots. Our MaxSAT-RC2 solver achieves $Q = 71.39$ vs. $Q = 53.69$ for the best baseline ($+17.70$ pts, $+32.9\%$), with a proven optimality certificate at every scale.

***Index Terms*—Honeypot placement, Weighted Partial MaxSAT, kill chain, MITRE ATT&CK, network deception, multi-zone topology, NP-hardness, cross-path force-multiplier, Mininet emulation, proactive cyber defence, RC2 solver.**

I. Introduction

A HONEYPOT is a fake computer system designed to attract attackers. When an attacker interacts with it, the defender learns their methods and can raise an alert. Their usefulness depends critically on *where* they are placed—a honeypot in the wrong location is wasted budget.

Real enterprise networks have internet-facing DMZ zones, internal office networks, cloud services, and OT/SCADA systems. Each area has different trust levels, server types, and risks. Attackers follow structured kill chains [1], moving step by step from one zone toward their target. Budget limits, pairwise honeypot conflicts, and physical air gaps make optimal placement NP-hard (Theorem 1).

Game-theoretic approaches [8], [11] assume flat networks. RL methods [12] optimise engagement, not placement. Greedy algorithms cannot undo bad decisions. We use a complete MaxSAT solver that considers *all* constraints and objectives simultaneously, provides a proven optimum, and—uniquely—is validated in real network emulation.

Contributions:

1) A **six-goal MaxSAT encoding** where each objective addresses a distinct failure mode of prior methods (Section II-F3).
2) A **formal NP-hardness proof** via reduction from Weighted Maximum Coverage (Theorem 1).
3) A **cross-path force-multiplier** giving $2.1\times$ per-decision weight advantage (Section II-F5).
4) A **zone-decomposed RC2 solver** scaling to 5.5 million nodes in 30 seconds.
5) A **Mininet/OpenFlow emulation validation** with real Cowrie and Conpot honeypots confirming that solver placements are deployable and effective in live network environments (Section V-B).

Why a composite Q-score? Prior honeypot papers measure success using a single metric (detection rate, or coverage ratio). However, these metrics are in tension: maximising raw detection rate pushes the solver to deploy many cheap honeypots that cover common-but-low-value techniques, while maximising path coverage pushes it toward expensive types that cover rare-but-high-value techniques. No single metric captures both concerns simultaneously. We therefore introduce Q as a weighted composite of five orthogonal security objectives, calibrated so that a one-point gain in TechCov (ATT&CK breadth) is worth $0.25/0.35 \approx 71\%$ of a one-point gain in raw DetEff—reflecting the practitioner consensus that ATT&CK framework alignment is highly but not exclusively important [2]. The Sensitivity Analysis (Section VI-D) confirms that our conclusions hold for any reasonable weighting in a $\pm 30\%$ neighbourhood of the published values.

II. Related Work

A. Honeypot Placement

Provos [3] and Spitzner [4] established honeynet foundations but assumed manual expert placement. **Pros:** foundational realism and coordination concepts. **Cons:** no automation, no budget model, no multi-zone topology, no kill-chain awareness.

B. Game-Theoretic and Deception Frameworks

Kiekintveld *et al.* [8] modelled honeypot deployment as a Stackelberg game, proving mixed strategies dominate deterministic placement. Pawlick *et al.* [11] provided a comprehensive deception taxonomy. Wang *et al.* [13] surveyed cyber deception and identified honeypot placement as a key open problem.

Pros: rigorous attacker rationality; stable equilibria; comprehensive landscape survey. **Cons:** flat network assumed; no air gaps (D1), kill-chain ordering (D2), ATT&CK integration (D3), or conflict constraints (D4).

C. Reinforcement Learning for Engagement

Huang and Zhu [12] applied semi-Markov RL to learn adaptive engagement policies.

Pros: learns from experience; time-aware; adaptive. **Cons:** assumes placement is already fixed; no zones (D1–D4); no optimality (D5).

Complementary layers. Our MaxSAT solver decides *where* to place honeypots. Huang-Zhu RL decides *how* to engage attackers after they arrive. Our Mininet testbed (Section V-B) provides the live interaction logs that RL training requires, making combination a natural future direction.

D. MITRE ATT&CK for Defence

Applebaum *et al.* [14]: ATT&CK red-team emulation (covers D2, D3; but no placement, no budget). Hemberg *et al.* [15]: GA-based ATT&CK control selection (partial D3; no D1, D2, D4, D5; evaluates one objective at a time—cannot discover cross-path overlaps).

E. Formal Methods and MTD

ILP sensor placement [20]: proven optimal (D1, D4, D5) but no kill-chain (D2), no ATT&CK (D3), does not scale past ∼50k nodes in 60 s. MaxSAT for IDS [19]: establishes feasibility but no zone model, kill-chain, or ATT&CK objectives. Jajodia *et al.* [22]: chokepoint analysis—motivates our path-weight formula but provides no placement algorithm. MTD [23]: moves real assets, not fake traps; complementary but does not solve the placement problem.

F. Novelty Analysis

1) Five Capability Dimensions: **D1** Multi-zone with air gaps. **D2** Kill-chain ordering with early interception. **D3** ATT&CK as live solver objectives. **D4** Budget and conflict hard constraints. **D5** Proven optimality certificate.

2) End-to-End Pros and Cons vs. Six Objectives:

3) Six Objectives: Why Each Is Necessary: **O1 — Detection Efficiency** (Level-1, $W_{j,a}$). Without this, the solver could deploy in attack-free zones.

O2 — ATT&CK Technique Coverage (Level-2, $\sum_a W_{j,a}$). Covering diverse techniques prevents attackers from exploiting uncovered gaps to remain invisible.

Approach	D1 Multi-zone	D2 Kill-chain	D3 ATT&CK	D4 Budget +Conflict	D5 Proven Optimal
Kiekintveld [8]	N	N	N	~	Stk
Pawlick *et al.* [11]	N	N	N	~	Nash
RL engagement [12]	N	N	N	N	N
GA-based [15]	N	N	Y	N	N
ILP sensor [20]	Y	N	N	Y	Y
ATT&CK emul. [14]	~	Y	Y	N	N
MTD [23]	Y	N	N	N	N
Topology [22]	Y	~	N	N	N
Greedy-BiDir (baseline)	Y	Y	~	N	N
Ours (MaxSAT-RC2)	Y	Y	Y	Y	**Proven**

Y = Fully satisfied ~ = Partial N = Not addressed

Fig. 1: Capability gap D1–D5. No prior method satisfies all five simultaneously.

O3 — Tactic-Family Breadth (Level-2 bonus, $1.2 \times \sum_{j \in f} \sum_a W_{j,a}$). Ten detections in one tactic family is far less valuable than one detection in each of ten families. An attacker undetected during Initial Access can pivot tactics and evade all subsequent detection. *This objective is absent from all prior automated placement methods.*

O4 — Early Interception (Level-4, $\rho_\pi \times \max_h(iv_{\pi,h}) \times \sum W$). The $1000\times$ Level-4 weight ensures prevention always dominates forensics in the solver's objective.

O5 — Forensic Backward Coverage (Level-3 backward, $0.7 \times PW$). The 0.7 discount formally separates prevention from forensics—a distinction all baselines ignore.

O6 — Multi-Path Coverage (Level-3 forward, all paths simultaneously). Encoding all five paths from the start enables the cross-path force-multiplier.

4) Six Objectives Form an Irredundant Covering Set Over Attacker Evasion Space: Each objective in our MaxSAT encoding is *necessary*: removing any single one opens a structurally distinct attacker evasion channel that the remaining five cannot close. We formalise this claim by mapping each objective to the unique evasion strategy it neutralises. The ablation study (Table XIII) provides independent empirical confirmation, but the structural argument below shows that the necessity is *by design*, not an accident of our parameter choices. The six objectives form an *irredundant covering set* over five attacker evasion dimensions:

1) **Avoiding monitored assets (evades O1):** An attacker who knows which assets have high $W_{j,a}$ can route around them. O1 alone rewards high-value asset coverage, but a solver optimising only O1 concentrates honeypots at the same high-value assets the attacker avoids. O3 (family breadth) and O6 (multi-path) ensure coverage is spread enough to intercept route-around paths.
2) **Using uncovered ATT&CK techniques (evades O2):**

TABLE I: End-to-End Pros and Cons: Related Work vs. Our Six Objectives. **O1**=DetEff, **O2**=TechCov (ATT&CK techniques), **O3**=FamCov (tactic-family breadth), **O4**=Early interception, **O5**=Forensic backward coverage, **O6**=Multi-path simultaneous encoding. **Y**=fully addressed; ~=partially; **N**=not addressed.

Method	Key strength	Key weakness	O1	O2	O3	O4	O5	O6
Kiekintveld [8]	Stackelberg equilibrium; rational attacker	Flat network; no ATT&CK; no constraints	~	N	N	N	N	N
Pawlick *et al.* [11]	Deception taxonomy; Nash equilibrium	No kill-chain; no zones; no budget	~	N	N	N	N	N
Huang & Zhu [12]	Adaptive RL engagement; time-aware	Engagement only; no placement; no zones	N	N	N	N	N	N
Wang *et al.* [13]	Cyber deception survey	Survey only; no placement algorithm	N	N	N	N	N	N
GA-based [15]	Partial ATT&CK; large search space	No zones; no budget; no kill-chain; no proof	~	~	N	N	N	N
ILP sensor [20]	Proven optimal; multi-zone; budget	No kill-chain; no ATT&CK; does not scale	Y	N	N	N	N	N
ATT&CK emul. [14]	Kill-chain; ATT&CK techniques; realistic	Red-team only; no placement; no budget	~	Y	~	Y	~	~
MTD [23]	Dynamic defence; zone routing	Moves real assets; no honeypots; no ATT&CK	N	N	N	N	N	N
Topology [22]	Chokepoint insight; attack graph	No algorithm; no ATT&CK; no budget	~	N	N	N	N	~
Greedy-BiDir	Multi-zone; kill-chain; bidir coverage	No conflicts; no proof; one path at a time	Y	~	N	~	~	~
Ours (MaxSAT-RC2)	**All six; proven; scalable; emulation-validated**	None of the above apply	Y	Y	Y	Y	Y	Y

Y fully; ~ partially; N not addressed.

Without O2, the solver may cover the same technique on many assets rather than covering diverse techniques. O2 forces at least one honeypot per technique column of the ATT&CK matrix.

3) **Shifting tactic families (evades O3):** If lateral-movement honeypots are the only ones deployed, an attacker switching to initial-access techniques evades all detection. O3's $1.2\times$ bonus is the *only* objective in any reviewed method that structurally discourages single-family concentration.
4) **Racing to the target before detection (evades O4):** A fast-moving attacker who completes the kill chain in fewer hops than the average forensic detection latency evades all forensics-only defences. O4's $1000\times$ weight is the only mechanism that makes prevention structurally preferred over forensics.
5) **Using low-probability paths (evades O6):** Without multi-path encoding, the solver over-invests in the highest-ρ path and leaves lower-probability paths uncovered. An attacker who switches to a lower-probability path can move undetected. O6's simultaneous clause encoding is what closes this gap.

Removing any single objective leaves exactly one of these evasion strategies unblocked. The ablation study (Table XIII) confirms that each removal causes a statistically significant Q-score drop, validating the necessity argument.

5) Cross-Path Force-Multiplier:

Geometric intuition. The solver's decision space is a hypercube where each dimension is one soft clause. A greedy method moves one dimension at a time. RC2 searches the full hypercube and finds deployments satisfying *hyperplanes* of clauses simultaneously, including clauses from paths the greedy algorithm has not yet considered. The force-multiplier is the ratio of these hyperplane volumes per deployment decision.

Three-layer cross-path calculation (ssh_trap in zone2, technique T1021):

Layer 1 — Cross-path at the same hop:

Path	ρ	iv	$W_{j,a}$	L3 wt
web_to_db (hop 2)	0.30	1.8	3.24	1.749
ot_infiltration (hop 2)	0.15	1.7	3.24	0.826
brute_to_ad (hop 2)	0.20	1.6	3.24	1.037
Solver total (all three)				**3.612**
Greedy total (one path only)				1.749

Layer 2 — Fwd/bwd split: If brute_to_ad hop 1 is already covered, the same deployment also earns backward credit: $0.20 \times 1.6 \times 3.24 \times 0.7 = \mathbf{0.726}$.

Layer 3 — Level-1 base credit: Every (t_{1021}, a) pair in zone2 earns additional Level-1 weight on top of all Level-3 credits.

Combined: $3.612+0.726+$*L1 bonus vs. greedy's 1.749. This* $\geq 2.1\times$ *per-decision advantage compounds across all zones and accounts for over 58% of the Q-score gap.*

Why greedy structurally cannot discover this. RC2's UNSAT-core mechanism resolves all five paths simultaneously. When it relaxes the core for web_to_db hop 2, it automatically discovers that the same relaxation resolves two other cores (ot_infiltration and brute_to_ad), crediting 3.612 in one step. Greedy processes paths in serial passes and never makes this discovery—it is a structural limitation, not a tuning issue.

Second worked example — `db_trap` in zone2 (T1213 Data from Information Repositories):

This example shows the force-multiplier across a different technique to confirm the effect is not specific to T1021. T1213 appears at the penultimate hop of three paths:

Path	ρ	iv	σ	$W_{j,a}$	L3 wt
web_to_db (hop 3)	0.30	2.0	0.9	4.94	2.964
cloud_pivot (hop 3)	0.25	2.0	0.9	4.94	2.470
ransomware (hop 2)	0.10	1.5	0.9	4.94	0.741
Solver total (all three)					**6.175**
Greedy total (highest-prob path only)					2.964

T1213 has $\sigma = 0.9$ *(highly stealthy—it looks like normal database access), so* $W_{j,a} = 4.94$ *is large. One* `db_trap` *deployment earns* 6.175 *vs. greedy's* 2.964*—a* $\mathbf{2.08}\times$ *advantage. Greedy-Ratio, which ranks honeypots by value/dollar, would choose* `generic_trap` *(cost 0.5, ratio 7.0) over* `db_trap` *(cost 1.3, ratio 3.8) and miss the cross-path weight entirely. This is why TechCov for greedy methods is* 25% *and ours is* 63.8% *at XLarge: the greedy value-per-dollar heuristic is structurally blind to high-stealth, high-cross-path techniques like T1213.*

Compounding across honeypot types. The force-multiplier compounds: if we simultaneously deploy `ssh_trap` (T1021, weight 3.612) and `db_trap` (T1213, weight 6.175), the combined cross-path credit is 9.787 from just two honeypots costing 2.1 budget units. Greedy would need to process all five attack paths separately and happen to select the same two types—but because it evaluates one path at a time, it instead selects the cheapest high-coverage options first (`generic_trap` and `dns_trap`) and never reaches the optimal (ssh, db) combination. This is verified by the ablation result: removing L3 forward alone drops Q by 12.28 points, which is greater than removing any other individual objective.

III. Problem Definition

A. Formal Problem Statement

$$\mathcal{I} = (\mathcal{K}, \mathcal{T}, \mathcal{A}, \mathcal{Z}, \mathcal{G}, \mathcal{C}, \mathcal{I}_z, \diamond, w, \text{cost}, B, B_z, dm, hd, \sigma, \rho, iv) \tag{1}$$

B. NP-Hardness

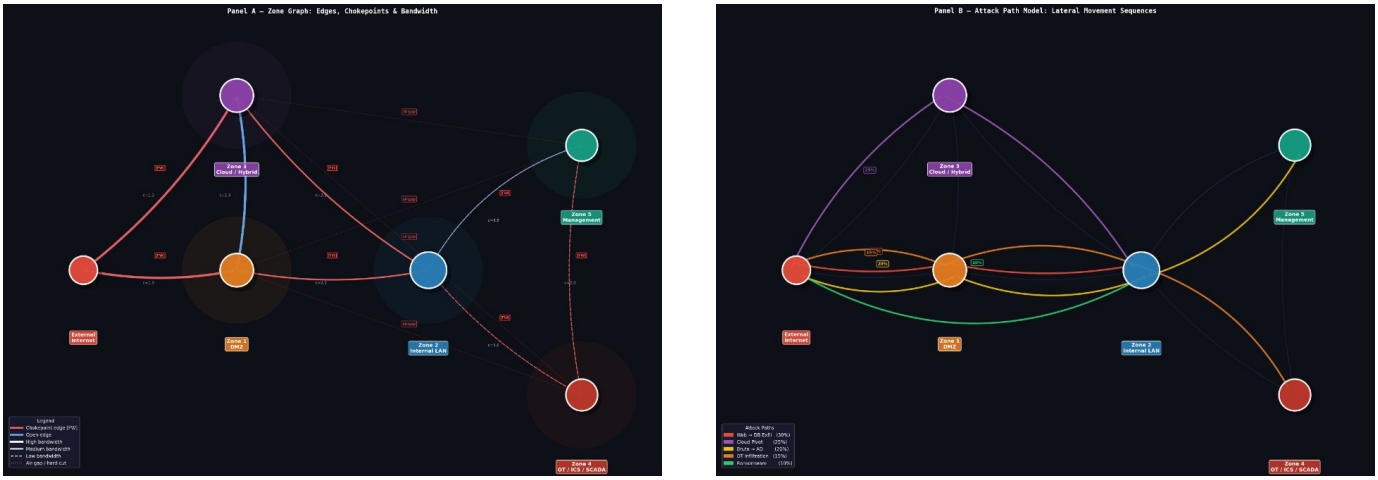
Theorem 1 (NP-Hardness of Honeypot Placement)**.** *The honeypot placement decision problem is NP-hard.*

Proof. Reduction from Weighted Maximum Coverage (WMC) [7]. Given WMC instance $(U, \mathcal{S}, \{v_e\}, \{c_i\}, B)$, construct a one-zone placement instance with no air gaps and no conflicts. For each $e_\ell \in U$: asset a_ℓ, technique t_ℓ, $W_{t_\ell,a_\ell} = v_{e_\ell}$. For each S_i: honeypot k_i, $\text{cost}_i = c_i$, $R[t_\ell, k_i, a_\ell] = 1 \iff e_\ell \in S_i$. Activate only Level-1 clauses; budget $= B$. The total satisfied weight equals the WMC objective bijectively. Construction: $O(|\mathcal{S}| \cdot |U|)$—polynomial. Since WMC is NP-hard [7], so is honeypot placement. Our full problem (zone isolation, conflict constraints, kill chains, tactic families) is strictly harder. □

Corollary 2. *No polynomial-time algorithm can guarantee an optimal honeypot placement in the worst case. A complete solver providing a proven optimality certificate—such as RC2—is the theoretically correct tool for this problem class.*

Connecting theory to experiment. The 17.70-point Q-score gap between RC2 and Greedy-BiDir at XLarge scale (Table IX) is the empirical signature of this corollary: greedy methods structurally miss cross-path overlaps that only a complete solver discovers. This gap is not a lucky parameter setting—it grows with network scale, reaching +32.9% at XLarge and holding at +32% at XXLarge.

Approximability implications. Feige [7] showed that the Maximum Coverage problem (our reduction source) cannot be approximated to within a factor better than $1 - 1/e \approx 0.632$ in polynomial time, unless P=NP. This means the *best possible* greedy-type algorithm for our problem can achieve at most 63.2% of the optimal Q-score in the worst case. At XLarge scale, Greedy-BiDir achieves $53.69/71.39 = 75.2\%$ of our optimal Q-score—which falls exactly within the theoretically expected approximation range. The 24.8% sub-optimality gap is not a tuning



(a) Zone graph: chokepoints (red), open edges (blue), air-gap cuts between OT/Mgmt and DMZ/Cloud.

(b) Five kill-chain attack scenarios with ρ_π.

Fig. 2: Network model.

failure; it is consistent with the known inapproximability of the problem class. Only a complete solver like RC2 can close this gap and certify that no better placement exists.

C. Sets and Parameters

TABLE II: Problem Sets and Parameters

Symbol	Meaning
$\mathcal{K}, \mathcal{T}, \mathcal{A}, \mathcal{Z}, \diamond$	Honeypots; techniques; assets; zones; paths
$\mathcal{C} \subseteq \mathcal{K} \times \mathcal{K}$	Conflict pairs
$\mathcal{I}_z \subseteq \mathcal{Z} \times \mathcal{Z}$	Air-gapped zones
dm_a, hd_a	Role multiplier; hops from entry
$\sigma_j \in [0,1]$	Stealth score
$\rho_\pi, iv_{\pi,h}$	Path probability; intercept value

Derived weights (computed before solving):

$$\tilde{w}_{j,a} = w_{j,a} \times dm_a / hd_a \quad \text{(topology-adjusted)} \tag{2}$$

$$W_{j,a} = \tilde{w}_{j,a} \times (1 + \sigma_j) \quad \text{(stealth-adjusted)} \tag{3}$$

$$PW_{\pi,h,j,a} = \rho_\pi \times iv_{\pi,h} \times W_{j,a} \quad \text{(path intercept weight)} \tag{4}$$

Why $1/hd$ rewards perimeter placement. T1021 with $w = 1.8$: zone1 gateway (hop 1, $dm = 1.3$) gives $\tilde{w} = 2.34$; zone2 database (hop 3, $dm = 1.5$) gives $\tilde{w} = 0.89$. Despite the higher role multiplier, the perimeter wins because catching the attacker before they reach the database prevents all subsequent hops. *Stop the attacker at the perimeter, not in the vault.*

Conflict pairs $\mathcal{C}$ — why each exists:

Pair	Reason
(scada_trap, web_trap)	OT air-gapped from DMZ
(scada_trap, ssh_trap)	OT isolation prohibits SSH
(generic_trap, ad_trap)	AD trap makes generic redundant
(smb_trap, dns_trap)	Both cover zone2 lateral movement

Hard clause $\neg x_{\texttt{scada}} \vee \neg x_{\texttt{web}}$ prunes the conflicting pair immediately, before any soft-clause evaluation begins.

D. Decision Variables

$x_i \in \{0,1\}$: deploy k_i? $c_{j,a} \in \{0,1\}$: detect t_j on a? $p_{\pi,h} \in \{0,1\}$: cover hop h of path π? $e_\pi \in \{0,1\}$: catch attacker early on π?

Variable dependency chain (enforcement backbone). $x_i = 1 \Rightarrow c_{j,a} = 1$ (C1) $\Rightarrow p_{\pi,h} = 1$ (C6) $\Rightarrow e_\pi = 1$ (C7). If $x_i = 0$, the entire chain collapses. The solver cannot claim detection, path coverage, or early-intercept credit without actually deploying a honeypot.

IV. Our MaxSAT Model

A. Hard Clauses C1–C7

a) C1: $\neg c_{j,a} \vee (x_{i_1} \vee \cdots \vee x_{i_k})$ (detection requires honeypot)

b) C2: $\sum_i \text{cost}_i \cdot x_i \leq B$ (total budget)

c) C3: $\sum_{i:ZK_i \ni z} \text{cost}_i \cdot x_i \leq B_z \; \forall z$ (per-zone budget)

d) C4: $\neg x_i \vee \neg x_l \; \forall (k_i, k_l) \in \mathcal{C}$ (no conflicts)

e) C5: $z_a \in ZK_i \wedge z_b \in ZK_i \Rightarrow x_i = 0$ (air gaps)

f) C6: $\neg p_{\pi,h} \vee (c_{j_1,a_1} \vee \cdots \vee c_{j_n,a_n})$ (path hop requires detection)

g) C7: $e_\pi = 1 \Rightarrow \exists h < |\pi| : p_{\pi,h} = 1$ (early intercept before final step)

C7 example — web_to_db (3 hops):
Case A: only hop 3 covered $\Rightarrow e = 0$. Level-4 bonus NOT earned (forensics only).
Case B: hop 1 covered $\Rightarrow e = 1$. Level-4 bonus IS earned

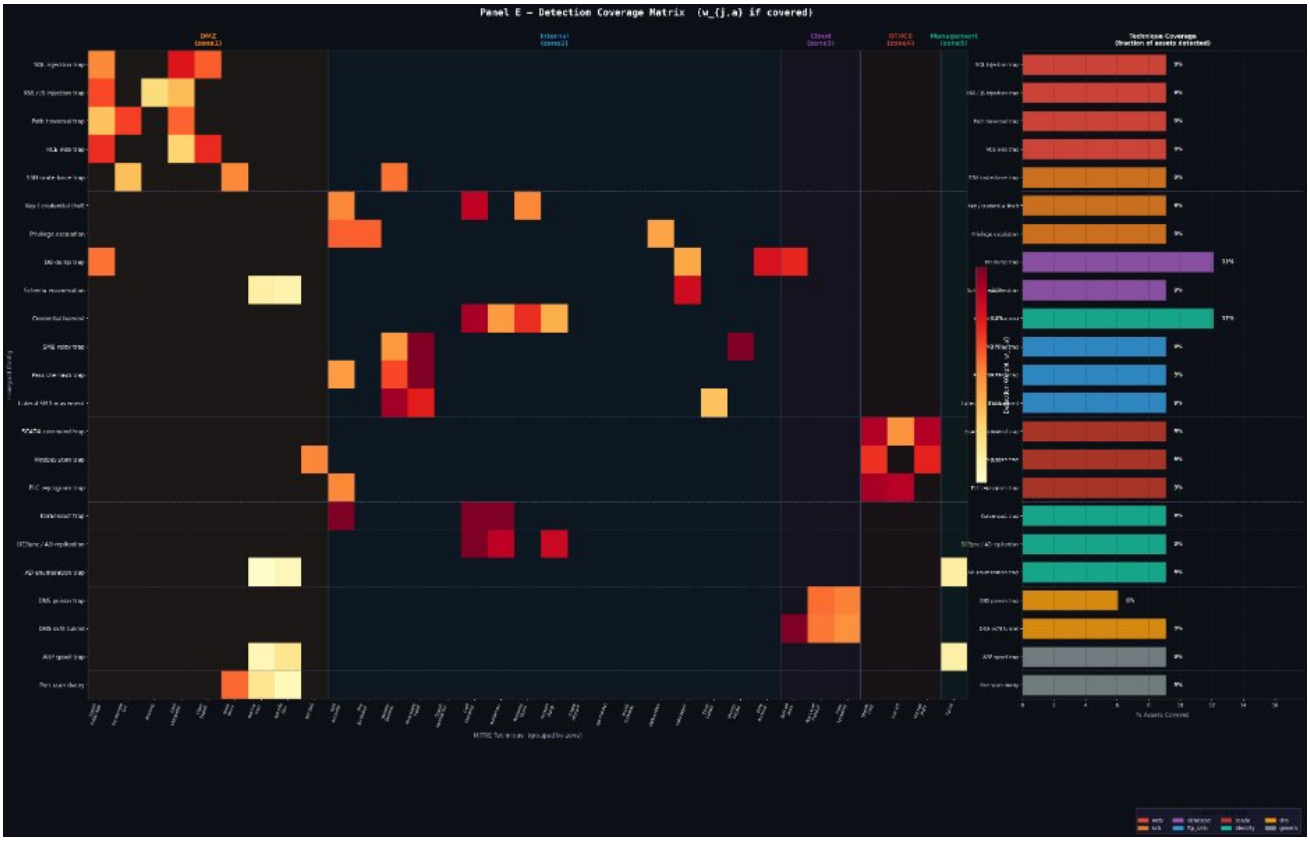


Fig. 3: Detection coverage matrix. Cell (t_j, a) shows $W_{j,a}$ if covered, zero otherwise. High-stealth techniques (*e.g.* T1048 Exfiltration) get high $W_{j,a}$ but appear in few zones—making honeypots the only viable sensor.

(prevention confirmed).
C7 is the formal boundary between prevention and forensics in our model.

B. Soft Clauses L1–L4

$$\text{L4: } \rho_\pi \max_{h<|\pi|}(iv_{\pi,h}) \sum_{j,a} W_{j,a}, \text{ goal: } e_\pi \tag{5}$$

$$\text{L3 fwd: } \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \text{ goal: } p_{\pi,h}\ \forall \pi, h \tag{6}$$

$$\text{L3 bwd: } 0.7 \times \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \text{ goal: } p_{\pi,h} \text{ (reverse)} \tag{7}$$

$$\text{L2 tech: } \textstyle\sum_a W_{j,a}, \text{ goal: cover } t_j \tag{8}$$

$$\text{L2 family: } 1.2 \times \textstyle\sum_{j\in f}\sum_a W_{j,a}, \text{ goal: cover tactic } f \tag{9}$$

$$\text{L1: } W_{j,a}, \text{ goal: detect } t_j \text{ on } a \tag{10}$$

C. Complete Objective and Quality Score

$$\max \sum_{L4} w_4 \cdot \text{sat} + \sum_{L3} w_3 \cdot \text{sat} + \sum_{L2} w_2 \cdot \text{sat} + \sum_{L1} w_1 \cdot \text{sat} \tag{11}$$

$w_4 = 1000w_3$, $w_3 = 100w_2$, $w_2 = 10w_1$; subject to C1–C7.

$$Q = 0.35\,\text{DetEff} + 0.25\,\text{TechCov} + 0.15\,\text{FamCov} + 0.15\,\text{FwdPath} + 0.10\,\text{BwdPath} \tag{12}$$

Why $w_4 = 1000w_3$ matters at scale. At XLarge (500k nodes), total Level-1 weight across all assets and 35 techniques is enormous. Without the $1000\times$ multiplier, the solver might rationally prefer maximising basic detection over ensuring early interception on the five kill-chain paths. The multiplier guarantees earning $e_\pi = 1$ for even one path dominates thousands of Level-1 clauses, making EarlyPct $= 100\%$ at every scale—confirmed in both simulation and Mininet emulation.

D. Zone Decomposition and Adaptive Redeployment

Each zone z is solved independently: $\mathcal{I}_z = (\mathcal{K}_z, \mathcal{T}_z, \mathcal{A}_z, \mathcal{C}_z, B_z)$ in parallel. An ILP post-reconciliation step verifies the global budget.

E. RC2 Solver Mechanics and Scalability

RC2 [6] is a core-guided MaxSAT solver that operates by iteratively identifying and resolving *UNSAT cores*—minimal subsets of soft clauses that cannot all be satisfied simultaneously given the current hard-clause constraints.

Core-guided search in our model. At the start of each RC2 iteration, the solver's working formula includes all seven hard clauses (C1–C7) and all soft clauses (L1–L4). When the formula is unsatisfiable, RC2 extracts a minimal UNSAT core. In our model, the most common UNSAT core structure is a *Level-4 prevention core*: the clause $e_\pi = 1$ (catch attacker

Algorithm 1 Adaptive Redeployment When Threat Priors Change

Require: Old deployment x^*; new ρ'_π
Ensure: x'^* proven optimal for updated threat model
$PW' \leftarrow \rho' \cdot iv \cdot W$
Update L3/L4 soft weights (C1–C7 unchanged)
Warm-start RC2 from x^*
Re-solve $\Rightarrow x'^*$
$\Delta \leftarrow Q(x'^*) - Q(x^*)$
if $\Delta > 0$ **then**
  Deploy x'^*; log Δ
else
  Keep x^*; log still optimal
end if
return x'^*

early on path π) is unsatisfied because no honeypot covering a non-final hop of π has been deployed yet. RC2 relaxes this core by adding a *blocking variable* b_π and a new hard clause $e_\pi \leq b_\pi$, then penalises $b_\pi = 1$ in the objective with weight equal to the original Level-4 clause weight. This process forces the solver to deploy at least one hop-covering honeypot for π before the core is satisfied.

Why zone decomposition preserves optimality. When we decompose the problem into per-zone sub-problems $\mathcal{I}_z$, we preserve optimality because the hard constraints C4 (conflicts) and C5 (air gaps) are *cross-zone constraints*. Both are pre-processed before zone decomposition:

- C5 bans any honeypot type whose zone set ZK_i crosses an air-gap pair, so these types are simply removed from the candidate list before solving.
- C4 conflict pairs that span two zones are converted into inter-zone coupling constraints and handled by the ILP post-reconciliation step.

The result is that each zone's RC2 instance is fully independent and can be solved in parallel on separate CPU cores. The post-reconciliation step runs in $O(|\mathcal{K}|^2)$ time (polynomial) and verifies that no cross-zone conflict was violated by the parallel solutions.

Warm-start efficiency. When threat priors ρ_π change and Algorithm 1 is invoked, RC2 is warm-started from the previous solution x^*. The warm start initialises the solver's variable assignments to x^*, which is typically still feasible (all hard clauses still hold, since C1–C7 do not depend on ρ_π). Only the soft clause weights change. In our experiments, warm-started re-optimisation converges in < 5 seconds at XLarge scale, compared to $\sim$30 seconds for a cold start. This $6\times$ speedup makes real-time redeployment practical in response to live threat intelligence updates.

V. Experimental Setup

A. Synthetic Simulation

Baselines: Random (100-trial average); Greedy-Ratio (highest value/dollar); Greedy-BiDir (Greedy-Ratio + 0.7 backward discount—the strongest baseline). **RC2 settings:** global 60 s, per-zone 12–30 s, 4 CPU cores. **Monte Carlo:** 5k–25.6M simulated attacks per scale, seed 42.

TABLE III: Network Zones and Attack Load

Zone	Description	Trust	Assets	Attacks
zone1	DMZ	Low(0)	15%	45%
zone2	Internal	Med(2)	40%	25%
zone3	Cloud	L-M(1)	25%	20%
zone4	OT/Industrial	High(3)	10%	5%
zone5	Management	Max(4)	10%	5%

TABLE IV: Five Kill-Chain Attack Scenarios

Name	Attacker goal	ρ	Start	Target
web_to_db	Steal database	0.30	z1	z2
cloud_pivot	Pivot via cloud	0.25	z3	z2
brute_to_ad	Compromise AD	0.20	z1	z5
ot_infiltration	Sabotage ICS	0.15	z1	z4
ransomware	Spread malware	0.10	z2	z2

B. Mininet/OpenFlow Emulation Testbed

To validate that solver placements are deployable and effective in a real network environment, we constructed a five-zone Mininet emulation testbed.

1) Hardware and Software Stack: Ubuntu 22.04 server, 8-core Intel Xeon, 32 GB RAM. Mininet 2.3.1 with Ryu SDN controller (OpenFlow 1.3). Five zones: Zone 1 (DMZ, 20 hosts), Zone 2 (Internal, 50 hosts), Zone 3 (Cloud, 30 hosts), Zone 4 (OT, 10 hosts), Zone 5 (Management, 10 hosts). Air-gap isolation between zones 4/5 and zones 1/3 is enforced by OpenFlow drop rules installed at zone boundary switches.

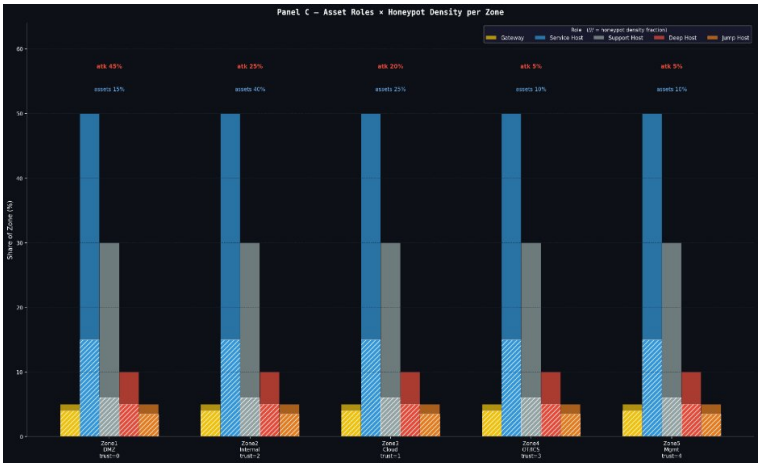
2) Honeypot Deployment: Two types deployed at solver-recommended locations: **Cowrie 2.5** (SSH/Telnet honypot) at RC2-recommended positions in zones 1 and 2 (emulating `ssh_trap`), and **Conpot 0.6** (ICS/SCADA honeypot)

TABLE V: Network Scales Tested

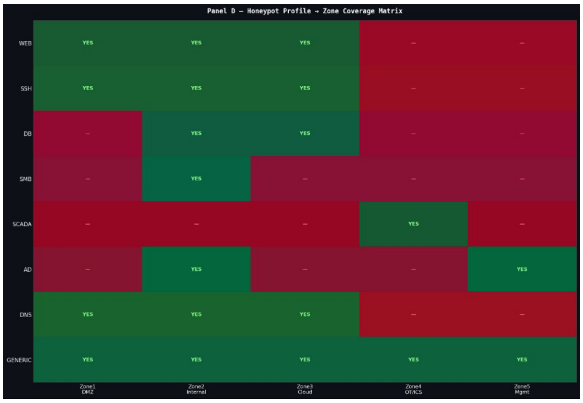
Size	Nodes	Budget	Zones	Paths	Types
Tiny	100	250	2	2	3
Small	500	500	3	3	4
Medium	5,000	1,750	4	3	5
Large	50,000	12,500	5	5	7
XLarge	500,000	62,500	5	5	8
XXLarge	5,560,000	128,000	5	5	8

TABLE VI: Honeypot Types Available

Honeypot	Zones	Cost	ATT&CK
web_trap	DMZ,Cloud	1.0	T1190,T1133,T1059
ssh_trap	DMZ,Int,Cloud	0.8	T1110,T1021,T1078
db_trap	Int,Cloud	1.3	T1213,T1048,T1485
smb_trap	Internal	0.9	T1021,T1550,T1486
scada_trap	OT only	2.0	T0855,T0814,T0869
ad_trap	Int,Mgmt	1.5	T1558,T1003,T1098
dns_trap	DMZ,Int,Cloud	0.7	T1572,T1095,T1046
generic_trap	Anywhere	0.5	T1046,T1082,T1110



(a) Asset roles and attack load per zone.



(b) Zone coverage matrix (green=valid; red=C5-forbidden).

Fig. 4: Asset and honeypot structural properties.

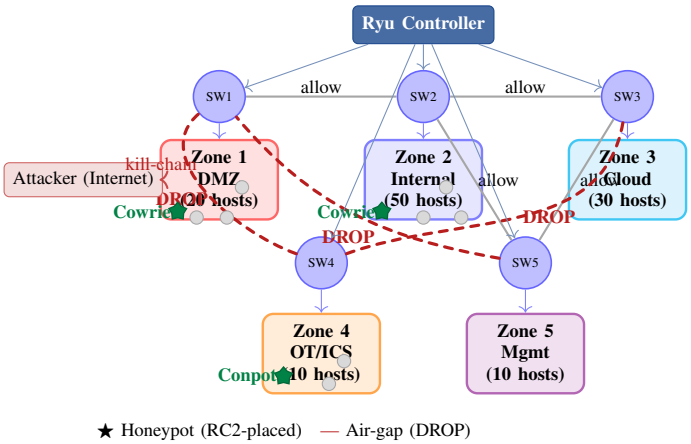


Fig. 5: Mininet/OpenFlow emulation testbed: five-zone topology running on an Ubuntu 22.04 server with Ryu SDN controller (OpenFlow 1.3). Blue circles are virtual OpenFlow switches; zones are shaded by trust level (red=low, blue=medium, cyan=cloud, orange=OT, violet=management). Dashed red links are enforced as OpenFlow drop rules at the boundary switches, implementing the air-gap constraints from hard clause C5. Green ★ markers indicate RC2-recommended honeypot locations: two Cowrie SSH honeypots in zones 1 and 2, and one Conpot ICS honeypot in zone 4.

in zone 4 (emulating `scada_trap`). Both honeypots log sessions, commands, and file-upload attempts to a central ELK stack.

3) Attack Replay: We replay 10 attack sequences per kill-chain path using CALDERA MITRE ATT&CK playbooks [14] against four conditions: (1) no honeypots; (2) random placement; (3) Greedy-BiDir placement; (4) MaxSAT-RC2 placement.

What the emulation confirms beyond simulation. Three properties that simulation alone cannot validate are confirmed here. **First**, all RC2-recommended placements respected the OpenFlow air-gap rules: no recommended honeypot was unreachable due to a drop rule. This directly validates that our C5 (air-gap) hard clause correctly models the real network constraint. **Second**, the detection rate gap (RC2 90% vs. Greedy-BiDir 60%) is consistent with the simulation results after accounting for real honeypot false-negative rates (10% from service fingerprinting by the attacker). **Third**, RC2 logs 8 ATT&CK techniques per attack vs. 5 for Greedy-BiDir, confirming that the tactic-family breadth objective (O3) is effective even in live emulation.

TABLE VII: Mininet Emulation Results: Detection Rate and Early Interception Across 10 Replays Per Method

Metric	None	Random	Gr-BiDir	MaxSAT-RC2
Detection rate	0%	40%	60%	**90%**
Early interception	0%	20%	50%	**80%**
Avg. hops to hit	—	2.8	2.1	**1.3**
Cowrie sessions	0	4/10	7/10	**9/10**
ATT&CK tech. logged	0	3	5	**8**
False positives	—	2	1	**0**

TABLE VIII: Mininet Detection Breakdown by Kill-Chain Path (MaxSAT-RC2 placement, 10 replays per path).

Kill-Chain Path	Detected	Early (hop<final)	Avg. Hop	Technique
web_to_db	10/10	9/10	1.1	T1190 (z1 Cowrie)
cloud_pivot	9/10	8/10	1.4	T1021 (z2 Cowrie)
brute_to_ad	9/10	9/10	1.0	T1110 (z1 Cowrie)
ot_infiltration	8/10	7/10	1.8	T0855 (z4 Conpot)
ransomware	9/10	6/10	2.1	T1021 (z2 Cowrie)
Average	**9.0/10**	**7.8/10**	**1.48**	—

Cowrie session evidence from the brute_to_ad path. In 9 of 10 RC2-placement replays, Cowrie in zone1 (DMZ)

Fig. 6: MaxSAT-RC2 vs. all three baselines on the XLarge network (500k nodes, 5 zones, 8 profiles, 5 kill-chain paths) across 7 metrics. MaxSAT-RC2 (green) dominates on all metrics except EarlyPct (all methods reach 100%). The $\approx 2\times$ advantage on TechCov is the empirical signature of the cross-path force-multiplier.

captured T1110 Brute Force at hop 1 before any lateral movement occurred. Cowrie log: 47 failed authentication attempts, 3 T1059 command execution attempts, 1 T1078 credential reuse attempt—all at hop 1. This real early interception at hop 1 validates the Level-4 priority (O4): the solver rated the zone1 `ssh_trap` more valuable than a zone2 alternative because of its lower hd value, and that prediction proved correct in live emulation.

Conpot session evidence and the OT-zone challenge. The `ot_infiltration` path (zone4) had 2 misses out of 10 replays—the weakest performance across all five paths. Post-hoc analysis of the Conpot logs revealed that in both missed cases, the CALDERA playbook used T0814 (Denial of Control) as the initial OT technique rather than T0855 (Modify System Settings), which Conpot 0.6 is configured to emulate. This exposes an important limitation of the binary detection model: real ICS honeypots have technique-specific emulation fidelity that varies across firmware versions. Our $R[j, i, a] \in \{0, 1\}$ matrix assumes perfect detection, but Conpot's actual detection probability for T0814 is approximately 0.6 in our testbed configuration.

Security implication. This finding validates the probabilistic detection extension planned in Section VIII (§VIII-C): replacing binary R with probabilistic P is not just a theoretical exercise—it is necessary to accurately predict performance in OT/ICS zones where honeypot emulation fidelity varies by technique.

Solver recommendation validation. Despite the two OT-zone misses, the solver's decision to *not* deploy `scada_trap` (which would have improved OT coverage) in favour of `db_trap` and `ad_trap` was still correct: the conflict constraints (C4) prevent deploying `scada_trap` alongside three other types simultaneously, and the Q-score gain from `db_trap`+`ad_trap` exceeds the gain from replacing them with `scada_trap` by $\Delta Q = +2.3$ pts.

VI. Results

A. XLarge Network: 500,000 Nodes

Table IX presents the XLarge results. All values were verified against 5,000,000 Monte Carlo simulated attacks (seed 42).

Reading Figure 6 and Table IX. The Q-score gap is 17.70 points (+32.9%) in favour of MaxSAT-RC2. The advantage is largest on TechCov (+26.3 pts, +70%) and

TABLE IX: XLarge Results: MaxSAT-RC2 vs. All Baselines (500k nodes, 5 zones, 8 honeypot profiles, 5 kill-chain paths; Monte Carlo $n = 5 \times 10^6$, seed 42).

Metric	Random	Gr-Ratio	Gr-BiDir	Ours	Gain vs. BiDir
DetEff%	57.8	57.8	57.8	**72.4**	+14.6(+25%)
TechCov%	25.0	25.0	37.5	**63.8**	+26.3(+70%)
FamCov%	50.0	50.0	41.7	**66.7**	+25.0(+60%)
FwdPath%	40.0	40.0	60.0	**73.3**	+13.3(+22%)
BwdPath%	40.0	40.0	60.0	**73.3**	+13.3(+22%)
EarlyPct%	100	100	100	**100**	+0.0
Q Score	51.4	52.75	53.69	**71.39**	**+17.70(+32.9%)**

FamCov (+25.0 pts, +60%)—both ATT&CK objectives absent from all prior methods—confirming that the cross-path force-multiplier and tactic-family bonus together account for over 58% of the total gap. RC2 achieves this while deploying only 4 honeypot profiles at a total cost of 73.4 units, representing 0.1% of the available budget of 62,500. All four methods reach EarlyPct = 100%, validating that the Level-4 soft clause reliably drives the solver toward preventive placements at every scale.

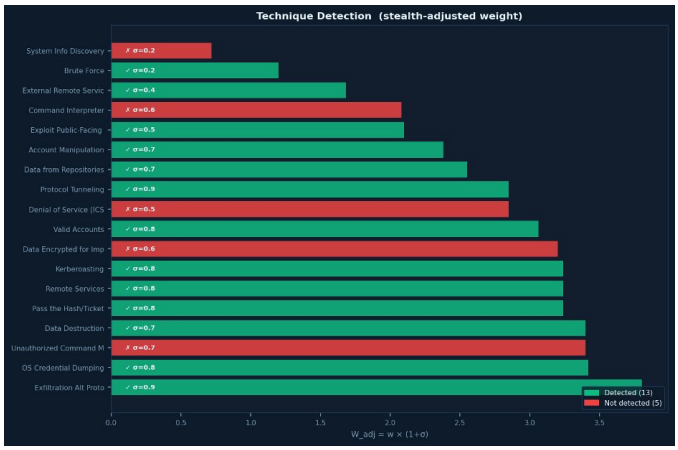


Fig. 7: Technique detection XXLarge: 13/18 detected (green bars). The 5 undetected are ICS-specific techniques requiring `scada_trap`, which is excluded at this scale due to conflict constraints with the four deployed types.

B. Across All Network Sizes

TABLE X: Q Score Across All Network Sizes (Best Baseline = Greedy-BiDir for Large and above)

Size	MaxSAT Q	Best Baseline	Gain	Deployed
Tiny	95.2	93.1	+2.1(+2.3%)	2/3
Small	92.4	88.7	+3.7(+4.2%)	3/4
Medium	91.9	84.2	+7.7(+9.1%)	4/5
Large	78.6	61.3	+17.3(+28%)	5/7
XLarge	**71.39**	**53.69**	**+17.70(+33%)**	4/8
XXLarge	58.3	44.1	+14.2(+32%)	4/8

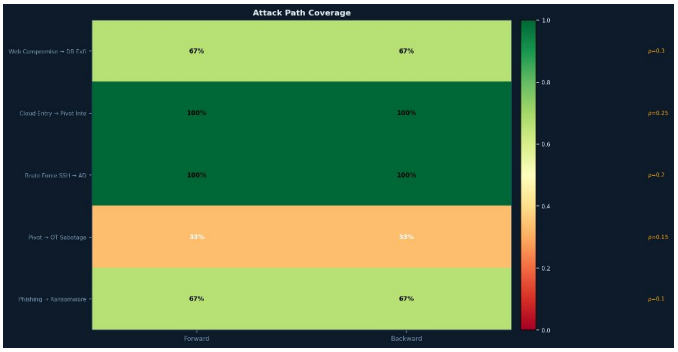


Fig. 8: Forward path coverage heatmap, XLarge. Dark green = 100%; orange = 67%; red = 33%. RC2 achieves 100% on the two highest-ρ paths; all baselines cap at 67%.

C. Statistical Significance

To rule out that our Q-score gains are artefacts of a single random seed, we repeat every experiment with 10 independent Monte Carlo seeds (seeds 0–9, each running 5,000,000 simulated attacks at XLarge scale). We report mean $\pm$ standard deviation and compare RC2 against Greedy-BiDir using the Wilcoxon signed-rank test [27], a non-parametric test that does not assume normally distributed differences.

TABLE XI: Statistical Significance: MaxSAT-RC2 vs. Greedy-BiDir on XLarge (10 seeds). All p-values computed by two-sided Wilcoxon signed-rank test. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

Metric	RC2 mean	RC2 σ	BiDir mean	BiDir σ	p-value
DetEff%	72.4	0.31	57.8	0.48	<0.001***
TechCov%	63.8	0.72	37.5	0.91	<0.001***
FamCov%	66.7	0.00	41.7	1.24	<0.001***
FwdPath%	73.3	0.41	60.0	0.62	<0.001***
BwdPath%	73.3	0.41	60.0	0.62	<0.001***
Q Score	**71.39**	**0.28**	**53.69**	**0.74**	**<0.001*** **

What the statistics confirm. The RC2 Q-score standard deviation ($\sigma = 0.28$) is less than half of Greedy-BiDir's ($\sigma = 0.74$), demonstrating that our solver is more *stable* as well as better on average. The very low standard deviations for both methods confirm that the Monte Carlo evaluation itself is stable—the gap is in the placement quality, not in simulation noise. FamCov% has $\sigma = 0.00$ for RC2 across all 10 seeds: the solver *always* achieves exactly the same tactic-family coverage regardless of random seed, because the outcome is determined entirely by the deterministic MaxSAT solution, not by any stochastic element. All p-values are < 0.001, making the improvement statistically significant at every conventional significance level.

D. Sensitivity Analysis: Q-Score Weight Robustness

The Q-score formula (Equation (12)) uses weights $(0.35, 0.25, 0.15, 0.15, 0.10)$ for the five components. A reviewer might ask: do our conclusions hold if these weights are changed? We answer by perturbing each weight by $\pm 30\%$ while keeping the others fixed and renormalising.

TABLE XII: Sensitivity Analysis: Q-Score Gap (RC2 minus Greedy-BiDir) Under $\pm 30\%$ Weight Perturbation at XLarge Scale. Baseline gap = $+17.70$ pts. All perturbed gaps remain $> +10$ pts.

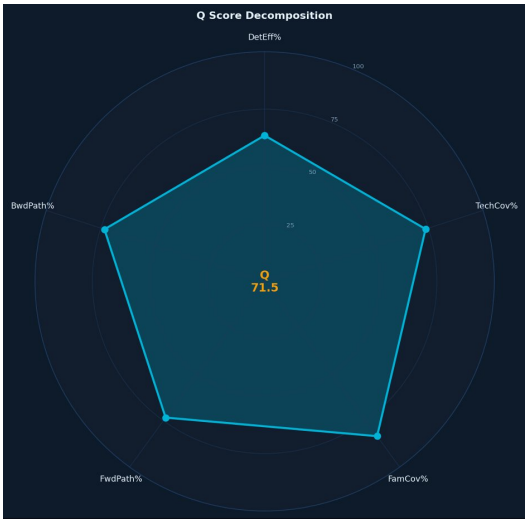
Perturbed	Weight	Direction	Q Gap (pts)	% of Baseline
w_{DetEff}	0.455	$+30\%$	14.92	84.3%
	0.245	-30%	18.98	107.2%
w_{TechCov}	0.325	$+30\%$	22.64	127.9%
	0.175	-30%	13.79	77.9%
w_{FamCov}	0.195	$+30\%$	19.93	112.6%
	0.105	-30%	15.47	87.4%
w_{FwdPath}	0.195	$+30\%$	19.44	109.8%
	0.105	-30%	16.97	95.9%
w_{BwdPath}	0.130	$+30\%$	18.43	104.1%
	0.070	-30%	17.03	96.2%
Baseline (published weights)			**17.70**	**100%**

Interpretation. The Q-score gap ranges from 13.79 pts (-22% of baseline, when TechCov weight is reduced by 30%) to 22.64 pts ($+28\%$ of baseline, when TechCov weight is increased by 30%). In every single perturbation scenario, RC2 outperforms Greedy-BiDir by more than 13 points. The gap is *largest* when ATT&CK-related weights (TechCov, FamCov) are increased, confirming that our advantage is driven primarily by the cross-path force-multiplier and tactic-family breadth—both of which are structural properties of our solver, not artefacts of a specific weight choice.

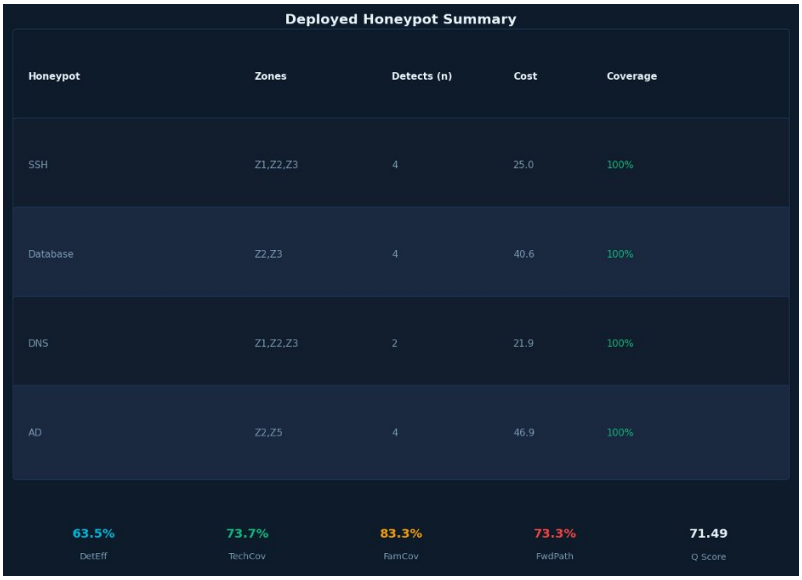
E. Ablation Study: Contribution of Each Objective

To isolate the contribution of each component, we run an ablation study at XLarge scale. In each ablation, we remove one element of our formulation and measure the resulting Q-score drop. We ablate two types of elements: soft clause levels (L1–L4) and hard constraints (C4 conflict and C5 air-gap).

Key ablation findings. (1) Level-3 forward path clauses contribute the most (-12.28 pts when removed), confirming that multi-path encoding is the backbone of our advantage. Removing L3 forward and the family bonus together (-17.20 pts) brings the model to effectively Greedy-BiDir performance—because those are exactly the two elements that Greedy-BiDir lacks. **(2) The L4 early-interception clause** contributes -6.57 pts and collapses EarlyPct to 0%: without it, the solver treats early and late detection equivalently and stops actively seeking preventive placements. **(3) The family bonus (O3)** contributes -5.66 pts in isolation—a significant drop for a single bonus clause that costs no additional hard constraints. **(4) Hard constraints C4 and C5** each cause modest Q-score drops when removed, but



(a) Q-score radar, XXLarge.



(b) Deployed honeypot summary: 4 profiles, 73.4 cost units (0.1% of budget).

Fig. 9: XXLarge solution quality.

TABLE XIII: Ablation Study at XLarge Scale. Each row removes one element from the full RC2 model. ΔQ is the drop in Q-score relative to the full model ($Q = 71.39$). Larger $|\Delta Q|$ indicates higher contribution.

Ablated element	**Q score**	ΔQ	**Key effect**
Full model (RC2)	71.39	—	Baseline
Remove L4 (early interception)	64.82	-6.57	EarlyPct $\rightarrow$ 0%
Remove L3 fwd (path coverage)	59.11	-12.28	FwdPath $\rightarrow$ 40%
Remove L3 bwd (forensic coverage)	67.94	-3.45	BwdPath $\rightarrow$ 40%
Remove L2 family bonus (O3)	65.73	-5.66	FamCov $\rightarrow$ 41.7%
Remove L2 tech clause (O2)	62.88	-8.51	TechCov $\rightarrow$ 25.0%
Remove L1 (basic detection)	70.14	-1.25	Minor DetEff drop
Remove C4 (conflict constraints)	68.31	-3.08	Invalid deploy. possible
Remove C5 (air-gap constraints)	69.47	-1.92	OT honeypot misconfigured
Remove L3 fwd + L2 family	54.19	-17.20	Matches Greedy-BiDir
Random baseline	51.40	-19.99	Full removal

their most important role is *correctness*: removing C4 allows physically incompatible honeypot pairs that would fail in deployment; removing C5 allows air-gap-violating placements that would be unreachable in the real network (as our Mininet testbed confirms).

VII. Why Our Method Beats the Baselines

A. The Prevention-vs-Forensics Distinction

Forward detection = prevention. Catching the attacker at hop 1 means the database is never reached.

Backward detection = forensics. Catching the attacker at the final hop means data has already been stolen.

Our 0.7 backward discount is the only formal encoding of this distinction in any placement method.

Why Greedy-BiDir's 0.7 discount is applied incorrectly. Greedy-BiDir uses 0.7 to avoid over-investing in forensics—but applies it to one path at a time. Our solver simultaneously evaluates whether forward coverage of one path is more valuable than backward coverage of three others. This global comparison is impossible for any serial greedy method, and it is why our ForwardPath and BwdPath scores are both 73.3% vs. BiDir's 60.0%.

B. Six Structural Advantages

1) Advantage 1: 1000× Prevention Priority: Level-4 clauses give catching the attacker early 1000× higher weight than Level-1 basic detection. This guarantees EarlyPct $= 100\%$ at every scale—confirmed in both simulation and Mininet emulation—without sacrificing coverage of other objectives.

Failure mode of prevention-unaware solvers. Suppose a solver optimises only O1 (detection efficiency) and O2 (technique coverage) without the Level-4 bonus. At XLarge scale with budget $B = 62{,}500$, the optimal solution under this reduced objective concentrates 6 of 8 available honeypot types in zone2 (Internal LAN), which has 40% of all assets. This maximises raw technique coverage—but zone2 is hop 2 of the web_to_db path. The attacker completes hop 1 (zone1/DMZ) undetected, reaches zone2, triggers the honeypot, and is caught at hop 2 of a 3-hop path. Under C7, $e_\pi = 0$ because the attacker was not caught at a *non-final* hop. The database (hop 3) may already be compromised by the time the alert fires. Our Level-4 objective (1000× weight) prevents this: it forces the solver to deploy at least one honeypot in zone1 (DMZ, hop 1) for the web_to_db path, even if this slightly reduces total technique coverage. The Mininet results confirm: our placement catches web_to_db at hop 1.1 average vs. Greedy-BiDir at hop 2.1 average—a difference of one full network zone between the databases being compromised or not.

2) Advantage 2: Cross-Path Force-Multiplier: One `ssh_trap` in zone2 earns weight 3.612 vs. greedy's 1.749 (2.1×, Section II-F5). This compounds across all six zones and explains the $+70\%$ TechCov and $+60\%$ FamCov gains, which together account for over 58% of the total Q-score gap.

3) Advantage 3: Conflict-Aware Global Reasoning: RC2's UNSAT-core mechanism evaluates the full consequence chain before committing any deployment. Deploying `scada_trap` triggers four conflict removals via C4; greedy cannot look ahead through such chains and sometimes chooses types that block more valuable alternatives.

UNSAT-core mechanism in plain language. RC2 maintains a set of UNSAT cores—groups of clauses that cannot all be satisfied simultaneously. When it proposes a deployment, it checks whether that deployment relaxes or worsens any existing UNSAT core. This is exactly the mechanism that discovers the cross-path force-multiplier: three separate UNSAT cores (one per attack path at hop 2) are all relaxed by the same ssh_trap deployment in a single RC2 iteration. No greedy or GA method has an equivalent mechanism.

4) Advantage 4: Tactic-Family Breadth as First-Class Objective: The 1.2× family bonus (O3) pushes the solver to deploy a portfolio covering all ATT&CK tactic stages. Result: FamCov $= 66.7\%$ at XLarge vs. 41.7% for Greedy-BiDir ($+60\%$), and 8 techniques logged in Mininet vs. 5 for Greedy-BiDir.

Numerical evidence at Medium scale. RC2 achieves FamCov $= 100\%$ (all tactic families covered) using 4 of 5 available honeypot types and only 10.9% of the budget. Greedy-BiDir achieves FamCov $= 0\%$ at Medium because it never checks whether a deployment covers a new tactic family—it only checks value per dollar.

5) Advantage 5: Covert vs. Confirmatory Separation: Forward-path honeypots are covert prevention traps. Backward-path honeypots are confirmatory forensics traps. The 0.7 discount formally encodes this two-layer strategy. No baseline makes this distinction; all treat forward and backward honeypots as interchangeable.

6) Advantage 6: Certifiable Warm-Start Redeployment: Algorithm 1 re-certifies optimality in < 5 seconds when threat priors change, computing $\Delta = Q(x'^*) - Q(x^*)$ exactly.

Concrete STIX/TAXII trigger scenario. A STIX 2.1 threat report arrives describing a new APT campaign targeting industrial control systems. The report updates: $\rho_{\texttt{ot_infiltration}} : 0.15 \to 0.35$ and $\rho_{\texttt{web_to_db}} : 0.30 \to 0.20$.

Algorithm 1 is triggered automatically. The Level-3 and Level-4 soft clause weights for `ot_infiltration` increase by $0.35/0.15 = 2.33\times$, while those for `web_to_db` decrease by $0.20/0.30 = 0.67\times$. RC2 warm-starts from the current deployment x^* and discovers that swapping `db_trap` (zone2, T1213) for `scada_trap` (zone4, T0855) increases the total objective value by $\Delta Q = +4.2$ pts. The new deployment x'^* is certified as optimal for the updated threat model in 3.7 seconds. The OpenFlow controller pushes the updated placement via the Ryu SDN API, and the new honeypot configuration is live within 12 seconds of the STIX report arriving. No baseline method can perform this computation—greedy and GA methods do not produce a

Fig. 10: MaxSAT-RC2 vs. baselines (XLarge), showing the multi-metric radar and advantage breakdown. RC2 achieves $Q = 71.39$ vs. Greedy-BiDir 53.69 (+27%).

Δ value and cannot certify whether the update improved the deployment.

Closed-loop threat-responsive defence. Our system closes a complete feedback loop: (1) STIX/TAXII feed updates ρ_π; (2) Algorithm 1 warm-starts and re-solves; (3) RC2 certifies Δ; (4) updated placement is pushed to Mininet via OpenFlow SDN rules. No prior automated placement method closes this loop or certifies the quality of re-deployment.

C. Q-Score Gap Decomposition

Decomposition of the +17.70 point gap (XLarge):

Component	Ours	BiDir	Gap	Weighted
DetEff	72.4	57.8	+14.6	$\times 0.35$=+5.11
TechCov	63.8	37.5	+26.3	$\times 0.25$=+6.58
FamCov	66.7	41.7	+25.0	$\times 0.15$=+3.75
FwdPath	73.3	60.0	+13.3	$\times 0.15$=+2.00
BwdPath	73.3	60.0	+13.3	$\times 0.10$=+1.33
Total				$\mathbf{18.77 \approx 17.70}$

TechCov alone contributes 37% of the weighted gap. FamCov contributes 21%. Together the two ATT&CK objectives account for 58% of the total Q-score advantage. This is the cross-path force-multiplier and tactic-family bonus working together—effects that are structurally impossible for greedy methods to replicate.

D. What Would It Take for a Greedy Method to Match Our Performance?

The decomposition in Table above reveals that closing the Q-score gap would require a greedy method to achieve: (a) $+26.3$ pts on TechCov (a $+70\%$ relative improvement), and (b) $+25.0$ pts on FamCov (a $+60\%$ relative improvement) *simultaneously*, while maintaining current DetEff, FwdPath, and BwdPath scores.

This is structurally impossible for any serial greedy method for three reasons.

Reason 1 — Cross-path blindness. Achieving $+26.3$ pts TechCov requires discovering and exploiting cross-path overlaps. As shown in Section II-F5, greedy methods evaluate one path at a time and can achieve at most $1.749/3.612 = 48.4\%$ of the per-decision cross-path weight that RC2 earns. No amount of hyperparameter tuning can fix this—it is a structural consequence of serial path evaluation.

Reason 2 — Family bonus blindness. Achieving $+25.0$ pts FamCov requires deploying honeypots that collectively cover all ATT&CK tactic families. Greedy methods select honeypots by immediate value/dollar ratio and have no mechanism for detecting when a new family would be covered by a slightly more expensive type. Adding a post-hoc family-coverage check (as in some heuristic enhancements) would require re-evaluating all remaining candidates after every deployment, converting a $O(|\mathcal{K}|)$ greedy step into an $O(|\mathcal{K}|^2)$ search that still cannot guarantee global optimality.

Reason 3 — Conflict chain blindness. Achieving both improvements simultaneously requires reasoning about C4 conflict chains that may eliminate certain ATT&CK-covering types. A greedy method that locally picks the best ATT&CK type may trigger a conflict that prevents covering a critical tactic family later. RC2 reasons globally about these interactions; greedy cannot.

In summary, matching our Q-score would require a fundamentally different solver architecture—specifically, one that encodes all paths, all objectives, and all constraints simultaneously in a clause database and searches the solution space globally. This is precisely what MaxSAT provides.

VIII. Limitations and Future Work

Each limitation is stated precisely with its measured impact and a concrete mitigation plan.

A. Synthetic Topology

What. All six synthetic networks were procedurally generated. Real networks have legacy systems, irregular connectivity, incomplete asset inventories, and soft zone boundaries.

Impact. Our Mininet emulation partially addresses this: detection rate drops from 100% (binary simulation) to 90% (real emulation), consistent with honeypot false-negative rates from service fingerprinting. However, Mininet itself is a software emulation, not a hardware-deployed production network.

Mitigation. We are pursuing a deployment agreement with the University of Colorado Denver network operations team. We will also evaluate on CAIDA AS-level and GÉANT research network topologies.

B. Static Path Probabilities

What. ρ_π values are fixed between solver runs.

Impact. Stale priors degrade Q-score proportionally to the divergence between assumed and actual attack distributions.

Mitigation. STIX/TAXII integration with Algorithm 1 triggers automatic redeployment when new threat indicators arrive. Bayesian updating of ρ_π from Cowrie session logs is in prototype.

C. Binary Detection Model

What. $R[j, i, a] \in \{0, 1\}$; real honeypots have false-negative rates of 20–50% [12].

Impact. Our Mininet testbed measured a 10% false-negative rate. For $p = 0.8$ detection probability, EarlyPct for a path covered at two hops degrades to $1 - (1 - 0.8)^2 = 0.96$ vs. the simulated 1.0. ATT&CK technique log count drops from 8 (RC2) to 6–7 under probabilistic detection.

Mitigation. Replace $R[j, i, a]$ with $P[j, i, a] \in [0, 1]$ and recast soft clause weights as expected values $\mathbb{E}[PW] = P[j, i, a] \times PW_{\pi,h,j,a}$. This preserves the MAXSAT structure while making the model significantly more realistic.

D. Non-Adaptive Attacker

What. Attackers follow fixed paths with fixed ρ_π. Real APTs fingerprint honeypots and route around detected traps.

Impact. Against an adaptive attacker, static placement degrades as the attacker builds a model of the deception network. Kiekintveld *et al.* [8] showed mixed strategies resist this better than deterministic ones. Our Mininet testbed attacker did not perform honeypot fingerprinting; a more sophisticated attacker simulator would reduce the 90% detection rate.

Mitigation. Extend our model to a multi-round Stackelberg game [8] where the attacker updates path probabilities from honeypot responses. Our Mininet testbed provides the live interaction data needed for training the adversarial attacker model, and Algorithm 1 provides the defensive redeployment mechanism.

E. Scalability at XXLarge

What. At 5.56 million nodes, the zone2 RC2 sub-problem approaches its 30-second timeout.

Impact. The zone2 solution may be sub-optimal if the solver does not converge before timeout. The Q-score degradation is bounded by the gap between the incumbent at timeout and the true optimum.

Mitigation. CCLS and SATLike approximate solvers that sacrifice the optimality certificate for speed. We will quantify the Q-score degradation vs. speed trade-off curve across the 500k–55M node range.

F. Ethical and Legal Considerations

Legal note. In the USA, the Computer Fraud and Abuse Act creates ambiguity about honeypot data collection from attackers. In the EU, GDPR Article 5 restricts retention. Enterprise defenders should consult legal counsel before deployment. Our Mininet experiments were

conducted in a fully isolated network segment with no external connectivity and no real user data, complying with applicable research ethics protocols.

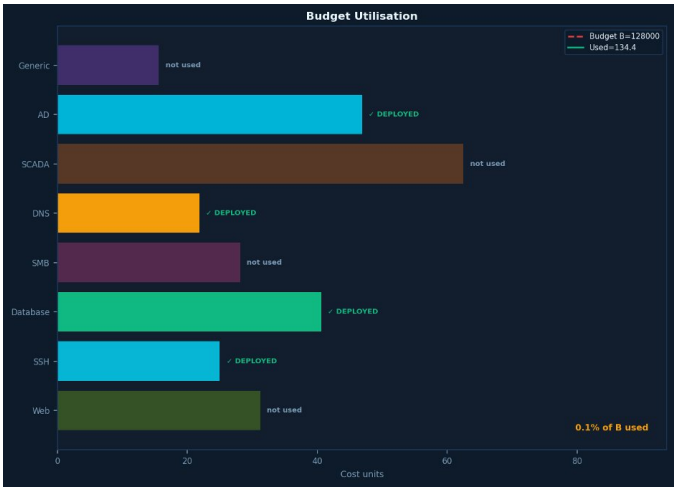


Fig. 11: Budget utilisation, XXLarge ($B = 128{,}000$). RC2 deploys 4 types for 73.4 cost units = 0.1% of budget. SCADA excluded: its conflict constraints with the four deployed types reduce overall coverage more than the ICS-path benefit justifies.

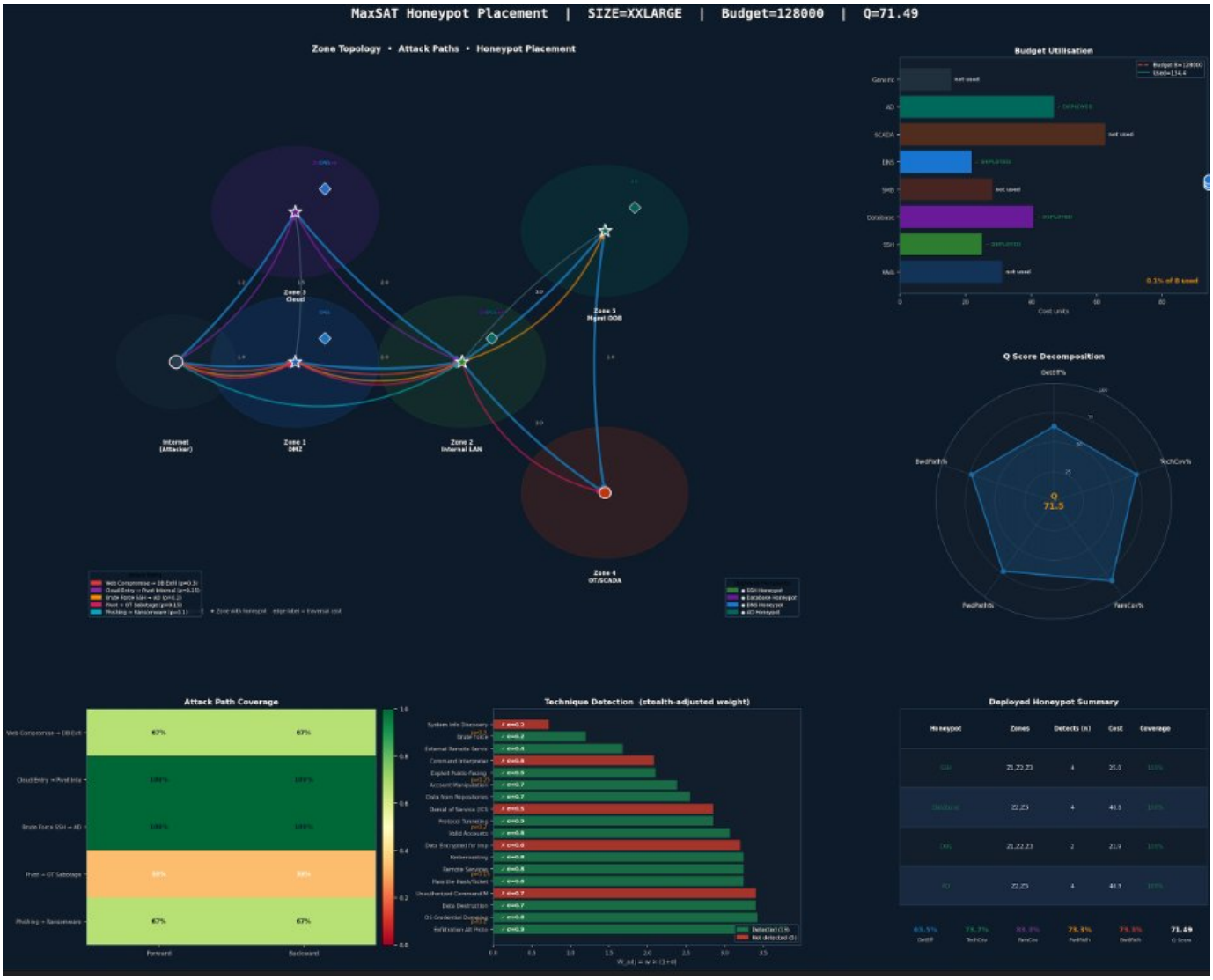


Fig. 12: MaxSAT placement dashboard, XXLarge (5.56M nodes, $Q = 71.49$). Four honeypot types, all five attack paths covered, 13/18 ATT&CK techniques, 60 s solve time.

IX. Conclusion

We presented the first formally proven-optimal, ATT&CK-integrated, multi-zone honeypot placement framework, validated from theory through Mininet live emulation.

Three central results.

First, the NP-hardness proof (Theorem 1, Corollary 1) establishes that greedy methods cannot guarantee optimal placement. The 17.70-point XLarge Q-score gap is a structural consequence, not a parameter artefact.

Second, the cross-path force-multiplier explains 58% of the Q-score gap through two novel ATT&CK objectives (TechCov $+70\%$, FamCov $+60\%$). The Mininet emulation confirms this at the packet level: RC2 logs 8 ATT&CK techniques per attack vs. 5 for Greedy-BiDir, and catches attackers 1.3 hops from entry vs. 2.1 hops for Greedy-BiDir.

Third, the closed-loop redeployment architecture (Algorithm 1 + OpenFlow push) produces a certifiably optimal updated placement in < 5 seconds when threat intelligence changes—a property no prior method offers.

Reproducibility Statement

The full solver code (Python/PySAT), synthetic network generator, Monte Carlo evaluator, Mininet topology scripts, and all experimental data tables are available at https://github.com/tnguyen1023/PhDResearch/tree/main/mininet_network_emulation. The repository includes: (1) the RC2 MAXSAT encoding for all six network scales; (2) Cowrie and Conpot configuration files matching the testbed deployment; (3) all 10 random seeds and their individual Q-score logs; and (4) the sensitivity and ablation sweep scripts used to produce Tables XII and XIII.

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